# Data flow Tracing

Initial state:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 0 & 5 \\ 7 & 8 & 6 \end{bmatrix}$$

Goal:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{bmatrix}$$

Heuristic:

$h(n) = 2$

$g(n) = 0$

$f(n) = 2$

Step - by - step Trace:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 0 & 5 \\ 7 & 8 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 3 \\ 4 & 2 & 5 \\ 7 & 8 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 5 \\ 7 & 0 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 7 & 8 & 6 \end{bmatrix} \leftarrow \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \\ 7 & 8 & 6 \end{bmatrix} \leftarrow \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \\ 7 & 8 & 6 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{bmatrix}$$

Path Built :

Initial → Right → Down → Goal

Visual diagram :

```
Initial (f = 2)
  ├ up (f = 4)
  ├ Down (f = 4)
  ├ Left (f = 4)
  └ Right (f = 2) ← chosen
      └ Down (f = 2) ← Goal
```